PS200C Causal Inference, Spring 2026

Introduction

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Modes of Statistical Inference

1. Descriptive Inference: summarizing and exploring data
   - means, correlations, regression coefficients
   - hypothesis testing
   - inferring “ideal points” from roll-call votes
   - inferring “topics” from texts and speeches

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    - predicting future state failures from past failures and covariates
    - predicting vote share from a survey of likely-voters
3. Causal Inference: engages *counterfactuals*
    - inferring whether the lack of war among democracies can be attributed to regime type
    - determining whether a medical treatment improved outcomes.
    - generally, estimating effect of some policy or institution or treatment on some outcome...

You cannot escape

Actually it's more than that,

1. *Some* descriptive enterprises are truly descriptive, but many are at least hoping the reader thinks the result is "suggestive" or "consistent with" a causal implication.

2. Issues such as missingness, case-selection, choosing what to condition on, and generalization all actually invoke causal assumptions, even if your goals are descriptive.

3. If you can do a randomized experiment, great, but many questions (e.g. mediation) aren't answered by the randomization.

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In fact, I hope the course helps to convince you that technical sophistication of methods often does not solve core problems...its not about the latest estimator, machine learning, or more complicated standard errors.

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Stylized illustration:

- you have data on X and Y
- you want to learn some effect of X on Y
- you compute something with data, say, $coef(Y \sim X)$
- trust me (for now): *without further assumption*, any causal effect can produce any coefficient...and thus any coefficient you get is equally suggestive of any causal effect.

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Curious as to what it takes? What could be better than data? Take the class!

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- ... it is not the method or estimator, much less software, that makes the result causal.

Causal Identification Requires Assumptions

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Credibility does NOT derive from "what method you used" or how new it is

- ...It is about finding a technique whose assumptions you can work with
- There are not "causal methods"! You can't say "it's causal because I did xyz."
- This is one of the hardest things for many students to absorb, in part because we are surrounded by mistakes and false advertizing.

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- We often cannot run randomized experiments or want to know the effect of something that "really happened". Can we still make a causal inference?
- How can we understand how vulnerable a given estimate might be to threats such as confounding?
- **How can you conduct top quality work that seeks to make causal claims in tricky situations, while accurately characterizing what we can and cannot be claimed?**

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- In lecture, the emphasis is on **intuition and assumptions**—truly understanding the (heroic) assumptions required by each identification strategy.
- You are expected and encouraged to **use AI tools** (Claude, ChatGPT, etc.) as tutors. We want you to learn to *think* about causal inference; the tools can help you *do* it.

Topics

#	Topic
1	Introduction and the causal inference problem
2	Potential outcomes and the identification problem
3	Randomized experiments
4	Selection on observables
5	DAGs and structural causal models
6	Sensitivity analysis
7	Difference-in-differences
8	Instrumental variables
9	Regression discontinuity
10	Special topics (time permitting)

Topics do not map exactly to weeks. Detailed readings for each topic will be posted on the course website.

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A note on coding and why I don't emphasize it.

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- A more comprehensive and annotated set of resources is on the course website's **Resources** page.

Course Requirements and Grades

- **In-class activities (20%)**
  - Short (5–10 min) pen-and-paper exercises at end of most lectures
  - Graded on completion and good-faith effort
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- **Final project (50%)**
  - Research paper ($\sim$10–15 pages) in groups of 1–3
  - Presentations in the last week of class
  - Due during finals week

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- Formulate a causal question, choose an identification strategy, carry out an analysis.
- Be honest about limitations—discuss which assumptions are most vulnerable.
- You may use AI tools freely for coding, data wrangling, and exploring ideas. Include a short section describing how you used AI in the research process.
- What AI *cannot* do for you: choose an identification strategy, evaluate whether its assumptions are credible, or interpret what results do and do not tell you. That is your job.

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- Learning to ask good questions of an AI is closely related to learning to think clearly about a problem.
- Each project milestone includes a brief **AI learning log**: one or two exchanges where you used an AI tool to clarify a concept, with a sentence about what you learned.
- This is not assessed for quality—it normalizes the practice and gives us insight into what topics are causing confusion.

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- Barney's office hours: Mondays 5 PM, Bunche 3244 (and by appointment)

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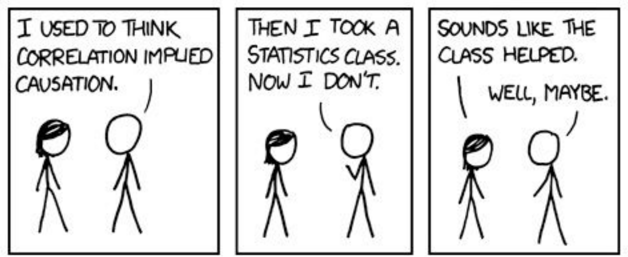
- Come to class, engage, ask questions. If you aren't going to attend regularly, don't take the class.

Some Resources

1. CAPS: https://www.counseling.ucla.edu. They also have a 24/7 crisis support at 310-825-0768.
2. The Bruin Fund offers funding for students with computing needs.
3. I'm always happy to help you get any services you need or to figure out what's available.



- Questions?
- Introductions!